

RKHS of $K(x, y) = (c + \langle x, y \rangle)^3$

1. Expansion via Binomial Theorem

Applying the binomial theorem with $p = 3$:

$$K(x, y) = (c + x^\top y)^3 = c^3 + 3c^2(x^\top y) + 3c(x^\top y)^2 + (x^\top y)^3.$$

Each power of $x^\top y$ lifts to a tensor inner product. Using the Frobenius / tensor inner product

$$\langle A, B \rangle_F = \text{tr}(A^\top B), \quad (x^\top y)^k = \langle x^{\otimes k}, y^{\otimes k} \rangle,$$

where $x^{\otimes k} = x \otimes \cdots \otimes x$ (k times) is a rank-1 symmetric tensor of order k .

2. Feature Map

The explicit feature map $\phi : \mathbb{R}^d \rightarrow \mathcal{H}$ stacks four tensor components:

$$\phi(x) = \begin{pmatrix} c^{3/2} \\ \sqrt{3c^2} x \\ \sqrt{3c} x^{\otimes 2} \\ x^{\otimes 3} \end{pmatrix},$$

where $x^{\otimes 2} = xx^\top \in \mathbb{R}^{d \times d}$ and $x^{\otimes 3} \in \mathbb{R}^{d \times d \times d}$, both flattened. The inner product recovers the kernel:

$$\langle \phi(x), \phi(y) \rangle = c^3 + 3c^2(x^\top y) + 3c(x^\top y)^2 + (x^\top y)^3 = K(x, y). \quad \checkmark$$

3. The RKHS as a Tensor Space

Every $f \in \mathcal{H}_K$ is parametrised by $(a, b, C, \mathcal{T})$ with

$$a \in \mathbb{R}, \quad b \in \mathbb{R}^d, \quad C \in \mathbb{R}^{d \times d}, \quad \mathcal{T} \in \mathbb{R}^{d \times d \times d},$$

and evaluated as:

$$f(x) = a + b^\top x + \text{tr}(C^\top xx^\top) + \langle \mathcal{T}, x^{\otimes 3} \rangle.$$

The cubic term contracts $\mathcal{T}$ against $x^{\otimes 3}$. Using the mode-1 matricisation $\mathcal{T}_{(1)} \in \mathbb{R}^{d \times d^2}$:

$$\langle \mathcal{T}, x^{\otimes 3} \rangle = x^\top \mathcal{T}_{(1)} (x \otimes x).$$

This yields a fully matrix-algebraic evaluation with no explicit index loops.

4. RKHS Inner Product

For $f \leftrightarrow (a, b, C, \mathcal{T})$ and $g \leftrightarrow (a', b', C', \mathcal{T}')$:

$$\langle f, g \rangle_{\mathcal{H}} = \frac{aa'}{c^3} + \frac{b^\top b'}{3c^2} + \frac{\langle C, C' \rangle_F}{3c} + \langle \mathcal{T}, \mathcal{T}' \rangle_F,$$

where the tensor Frobenius inner product is $\langle \mathcal{T}, \mathcal{T}' \rangle_F = \text{tr}(\mathcal{T}_{(1)}^\top \mathcal{T}'_{(1)})$. The RKHS norm is therefore:

$$\|f\|_{\mathcal{H}}^2 = \frac{a^2}{c^3} + \frac{\|b\|^2}{3c^2} + \frac{\|C\|_F^2}{3c} + \|\mathcal{T}\|_F^2.$$

5. Representer in Tensor Form

The kernel section $K(\cdot, x)$ —the representer of evaluation at x —corresponds to:

$$K(\cdot, x) \leftrightarrow (c^3, \quad 3c^2x, \quad 3cxx^\top, \quad x^{\otimes 3}).$$

The reproducing property is verified cleanly:

$$\begin{aligned} \langle f, K(\cdot, x) \rangle_{\mathcal{H}} &= \frac{a \cdot c^3}{c^3} + \frac{b^\top (3c^2x)}{3c^2} + \frac{\langle C, 3cxx^\top \rangle_F}{3c} + \langle \mathcal{T}, x^{\otimes 3} \rangle_F \\ &= a + b^\top x + x^\top Cx + \langle \mathcal{T}, x^{\otimes 3} \rangle = f(x). \quad \checkmark \end{aligned}$$

6. Kernel Matrix (No Loops)

On a dataset $X = [x_1 | \dots | x_n] \in \mathbb{R}^{d \times n}$, define the shifted Gram matrix $\mathbf{G} = c\mathbf{1}\mathbf{1}^\top + X^\top X$. The full kernel matrix is the Hadamard cube:

$$\mathbf{K} = \mathbf{G} \circ \mathbf{G} \circ \mathbf{G}.$$

No loops required—assembled from one BLAS matrix multiply and two elementwise products.

7. General Degree- p Pattern

The structure generalises to $K(x, y) = (c + x^\top y)^p$ via the binomial theorem:

$$K(x, y) = \sum_{k=0}^p \binom{p}{k} c^{p-k} \langle x^{\otimes k}, y^{\otimes k} \rangle.$$

Degree k	Feature	Weight in $\ f\ _{\mathcal{H}}^2$	Parameter
0	1	c^{-p}	$a \in \mathbb{R}$
1	x	$\left[\binom{p}{1}c^{p-1}\right]^{-1}$	$b \in \mathbb{R}^d$
2	$x^{\otimes 2}$	$\left[\binom{p}{2}c^{p-2}\right]^{-1}$	$C \in \mathbb{R}^{d \times d}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$x^{\otimes k}$	$\left[\binom{p}{k}c^{p-k}\right]^{-1}$	$\mathcal{T}^{(k)} \in \mathbb{R}^{d^k}$
p	$x^{\otimes p}$	1	$\mathcal{T}^{(p)} \in \mathbb{R}^{d^p}$

The RKHS is the space of **polynomials of degree** $\leq p$ on $\mathbb{R}^d$, with geometry determined entirely by symmetric tensor algebra and Frobenius inner products—no index gymnastics needed.